Essays

|!|BCEEA0220000001|!| ★★ **Cross topic**

John participates in a 100 m race at his school's Sports Day. Before the race, he drank a glucose-rich drink. Describe how glucose from the drink can be supplied to John's leg muscles to fuel his running. (10 marks)

##

Journey of glucose from John's alimentary canal to his leg muscles: (Max. 3)

- After the glucose-rich drink is swallowed, peristalsis of the alimentary canal moves the glucose from the oesophagus to the small intestine. (1)
- In the small intestine, glucose is absorbed into the blood capillaries inside villi by diffusion and active transport. (1)
- Glucose is carried by the blood and circulates around the body. (1)
- At the leg muscles, glucose diffuses from the blood to the muscle cells down the concentration gradient as glucose is constantly consumed by the cells. (1)

Leg muscles use glucose to release energy aerobically and anaerobically: (Max. 4)

- During vigorous exercise such as a 100 m sprint, leg muscles cannot obtain enough oxygen for aerobic respiration. The muscles carry out aerobic and anaerobic respiration at the same time. (1)
- In aerobic respiration, each molecule of glucose is completely broken down through glycolysis, the link reaction, the Krebs cycle and oxidative phosphorylation (1)
  with a yield of 38 molecules of ATP. (1)
- In anaerobic respiration, glucose goes through glycolysis only and is converted to pyruvate which enters the anaerobic pathway. (1)
- Without oxygen, NADH transfers its hydrogen to pyruvate, which is then reduced to lactic acid. NAD is regenerated to accept hydrogen in glycolysis once again. (1)
- The net ATP yield of anaerobic respiration in muscle cells is two molecules of ATP from glycolysis. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Applying concepts	7
Communication	3

© 2020 Aristo Educational Press Ltd.

|!|BCEEA0220000002|!| ★★ **Cross topic**

Photosynthesis and respiration are two important processes that support life. Describe how green plants convert light energy from the Sun to chemical energy in organic food, and how the green plants release energy in the form of ATP from organic food for cellular activities. (11 marks)

##

From light energy to chemical energy in organic food: (Max. 4)

- Green plants require light energy, carbon dioxide and water to produce glucose and oxygen. (1)
- Photosynthesis occurs in two stages. In photochemical reactions, light energy captured by chlorophyll molecules drive photophosphorylation and photolysis of water. (1)

  Photophosphorylation synthesizes ATP from ADP and phosphate, while photolysis of water produces NADPH and oxygen. (1)
- In the Calvin cycle, carbon dioxide combines with a 5-C carbon dioxide acceptor to form two molecules of a 3-C compound. The 3-C compound is reduced to triose phosphate. (1)

  Some of the triose phosphate formed is synthesized into glucose. Glucose may be used in respiration to release energy. (1)

Conversion of chemical energy in organic food to chemical energy in ATP: (Max. 4)

- In the presence of oxygen, glucose can be completely broken down to release energy. Photosynthesis can supply oxygen and glucose for respiration. (1)
- Glucose is broken down into triose phosphate and then pyruvate in glycolysis. Glycolysis synthesizes two molecules of ATP from one glucose molecule. (1)
- Pyruvate is converted to acetyl-CoA in the link reaction, with the release of carbon dioxide. Acetyl-CoA enters the Krebs cycle. (1)
- The Krebs cycle produces ATP, NADH, FADH and carbon dioxide. (1)
- When electrons from hydrogen atoms of NADH and FADH pass through a series of electron carriers in the inner membrane of mitochondrion, energy is released to phosphorylate ADP to ATP. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

© 2020 Aristo Educational Press Ltd.